A distinct 194-state Turing machine calibrated to the consistency of PA

Comparison of the complete supplied archive corpus,
a new composition, and the search below 194 states

Research memorandum prepared for Vladimir Reshetnikov

13 November 2026 (Pacific time)

Abstract

The strict state-reduction target was not attained in this investigation. The delivered alternative, PA194C, is a complete blank-start, binary, one-tape machine with 194 working controls, for which the mathematical argument establishes

$$\mathrm{PA} \vdash \big(\neg\,\mathrm{Halt}(M_{194C}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})\big).$$

It composes the 67-reader, seven-fold PA194 decoder with the previously calibrated Combined command controller. Six native transition rows change; the source has 3,649 primitive instruction sites instead of 3,673. The command behavior changes substantially: Shift tests the cached head alone, and successful modus ponens returns its conclusion in that cache. A fresh census of all 36 incoming archives and the direct PA tables finds no previous 194-state implementation of this controller, including under state relabeling and tape reflection. All 4,749 source branch contracts are checked symbolically for arbitrary registers, with a separate literal-tape implementation rederiving the required list lemmas. Exact blank initialization uses 57,922 transitions. The proof is an ordinary mathematical statement of PA-provability with executable supporting certificates; it is not a newly elaborated end-to-end Lean or Rocq proof. No optimality or lower-bound claim is made.

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1 Outcome, machine convention, and exact deliverable

The requested fallback is achieved. The new construction is PA194C, whose literal table is `construction/output/pa194C.tm`. Its conventions are: working states $0, \ldots, 193$, initial state 0, separate halt label 194, one two-way infinite tape over $\{0, 1\}$, all cells initially zero, and initial head coordinate zero. Each of the 388 rows writes one bit and moves exactly one cell left or right. There are no stationary moves, preloaded program symbols, extra tracks, or counted-away startup controls. Appendix A gives the entire transition function.

The authoritative dense table has SHA-256

```
d759313c03b515293fe06b12769fa1771d3829a2bad89febaa9b4391990f3a4b
```

Every working control is reachable in the directed transition graph. This graph property does not assert that every control occurs in the actual blank run.

Fix a conventional primitive-recursive proof predicate $\mathrm{Prf}_{\mathrm{PA}}$ for first-order PA with full induction and set

$$\mathrm{Con}(\mathrm{PA}) \equiv \neg\exists p \, \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner).$$

Let $\mathrm{Halt}(M_{194C}, \mathrm{blank})$ be the usual existential assertion that there is a finite valid transition trace from the specified blank configuration to the halt label. The principal claim is an internal biconditional of PA, not merely a metatheoretic statement conditional on standard-model soundness. Its proof composes syntactic refutation translations, a command compiler, and positive physical simulations. The detailed logical argument appears in Section 4. Actual nonhalting follows if PA is consistent; that consistency assumption is not added as an axiom when proving the biconditional.

1.1 What is new

The diagonal-vacuity calculus, the separated ten-entry emitter, the Combined controller, and the seven-reader fold technique are all credited prior components. The new result is their literal 194-state composition, its complete source and physical binding, its independent replay, and its comparison against the current corpus. Previously supplied 195C uses the Combined controller with a decoder costing 61 effective reader controls. The new decoder costs 60. Previously supplied PA194 tables use the original controller. Thus this is a new combination with an actual command-level difference, rather than an attempt to count an old saving twice.

2 Comparison corpus and substantive design families

All repository comparisons are pinned to ProveIt commit

```
768c2be1a8e4d2fcf8aa3443c7b0f084d145734c.
```

The inspected locations are the user's PA research subtree, `docs/incoming`, ZFC research subtree, and ConciseIPRA subtree [1, 2, 3, 9, 13]. The complete archive catalogue is in Appendix B; exact member locators, byte hashes, and graph classifications are in `comparison/`.

The fresh census descends into nested ZIP files, traversing each distinct archive payload once and retaining every archive reference. It recognizes complete numeric five-column binary tables in `.tm` and `.txt` members, and all direct PA `.tm` files. A recognized table must have both read-symbol transitions at every working state and exactly one target without outgoing rows. Arbitrary machine definitions embedded in source languages or JSON are outside this literal-table recognizer; the manuscript and README comparison supplements it. No incoming program is executed by the census.

Quantity	Observed count
Top-level incoming ZIP archives	36
Distinct recursively encountered ZIP payloads	49
Archive references, including repeated payloads	77
Maximum nested archive depth	8
Members in distinct archive payloads	26,258
Recognized complete table occurrences	2,519
Byte-distinct tables	606
Action-graph classes, disregarding the designated start	437
Direct literal PA `.tm` files	14

The 437 graph classes include ancestors, failed experiments, and negative controls. They are not 437 established consistency machines. Two disconnected small synthetic controls have no state reaching their full working graph and are excluded from that particular graph count. Graph comparison canonicalizes over every root reaching the full working graph. A mismatch therefore rules out a fixed-start relabeling; a match requires the actual initial-state convention before asserting semantic identity.

2.1 The main lineage

Family	Main structural change	States	Sites	Startup
Joint/Compatibility U209	Unary weak-cover source, before history fold	209	3,909	48,140
James Chang PA209	Different PA209 source and allocation	209	3,948	58,210
U208	Native 172/211 history fold, two original graph classes	208	3,909	48,140
PA196	Smaller native axiom program and physical allocation	196	3,812	58,154
Combined196	Cached Shift and successful MP result in H	196	3,788	57,922
Diagonal196	Formula-code diagonal vacuity, 191 AX items	196	3,700	58,154
Separate196	Separate diagonal head, 190 AX items	196	3,673	58,154
Original PA195	Diagonal reader allocation, $134 + 68 - 7$	195	3,700	58,154
195S	Separate reader allocation, $134 + 67 - 6$	195	3,673	58,154
195C	Separate emitter plus Combined commands	195	3,649	57,922
M195S8	Eight folds plus one extra scanner control	195	3,700	59,164
PA195DC	Original diagonal emitter plus Combined commands	195	3,676	57,922
PA194 (primary)	Separate emitter, $134 + 67 - 7$	194	3,673	58,154
PA194 (second)	Separate emitter, $134 + 68 - 8$	194	3,673	58,154
New PA194C	Primary 194 reader with Combined commands	194	3,649	57,922

Here “sites” means primitive source occurrences, not physical states; “startup” means the documented exact initialization length. Historical numbers describe their supplied certificates, whereas PA194C’s

numbers were freshly replayed in this investigation. The table condenses related families; the archive catalogue records every top-level packet.

The Joint/Compatibility U209 ancestor and James Chang's PA209 are distinct programs despite their equal state counts. U208 uses the native 172/211 history fold of its own U209 ancestor. Its first nine advertised tables represent two rooted graphs; later helper variations enlarge that family. PA196 changes the axiom program and allocation, so the U208 saving cannot simply be subtracted from PA196 again.

The major logical reduction in the later PA line replaces a more expensive vacuity gate by the diagonal instance

$$\Gamma(B) = B \to \forall x_B B.$$

Every label in the formula of code B is smaller than B, making x_B a fresh variable. The emitter decreases from 201 to 191 macro tokens. Keeping Γ as a separate stack entry removes one implication constructor, giving 190 tokens and a ten-entry rather than nine-entry AX interface. This changes theorem-selection indices and requires an adjusted command compiler. The physical association and sharing of this same word account for the later 195- and 194-state allocations.

2.2 Every existing 194-state class

The census contains seven 194-state byte forms belonging to four action graph classes. Three are supported written constructions; the fourth is explicitly rejected.

1. **Primary PA194.** It uses $134 + 67 - 7 = 194$ controls: 134 fixed controls, 67 raw AX readers, and seven compatible contextual folds. Its source has 3,673 sites and 4,780 branch contracts. The dense table begins with hash `4bfea62b17aad7d8` and is in the first PA194 archive [4].
2. **PA194_AD.** The first archive also exchanges physical A and D register positions, retaining the same logical source and command protocol. Its dense hash begins `27613aa92da79ba2`. This is a different physical graph but the same logical algorithm.
3. **Second PA194.** The second archive uses $134+68-8 = 194$ controls, the same 190-token word, and 3,673 sites. Its dense hash begins `9087d70fc0e1878c`. Its 32 local fold cases include a seven-step simulation [5]. One more raw reader enables two more compatible folds, so minimizing raw decoder nodes alone is the wrong objective.
4. **Rejected scanner-194.** The follow-up packet's native hash begins `c48d6711e9c73366`. Its local 202-to-194 quotient is valid, but its source contains a seven-zero address run which the old scanner mistakes for a separator. The first failed contract is source ordinal 1003, the zero branch on logical register 1. The repaired machine adds a scanner control and has 195 states. A valid local quotient of an invalid source implementation does not establish a consistency machine.

All four existing native 194-state graphs have the original six-row controller fingerprint. None uses the Combined command behavior. The comparison includes the rejected graph to avoid treating an earlier failed experiment as new.

2.3 Other lower-state candidates and proof layers

At 195 states the census has 82 occurrences, 25 byte forms, and 15 graph classes. Positive reported families include original PA195, separate-head 195S and 195C, diagonal Combined PA195DC, and repaired-scanner M195S8. Additional packets contain the rho-parameter variant PA195R, negative diagonal PA195N, bounded local-quotient outputs, and rejected diagnostics. Their distinct evidence status is retained in the catalogue, rather than classifying every 195-state file as a proved construction.

The direct PA tree's numerous `pa195-*` directories are mainly proof layers for existing tables. The quotient/dense/ordinary/zipper developments prove substantial ordinary run equivalences. Runtime, site binding, symbolic batches, and source iteration establish other parts of the compiler chain. The arithmetic endpoints still have machine-specific reduction obligations. In particular, the PA README explicitly does not claim the complete iPRA endpoint. This missing formalization is different from a counterexample to a mathematical proof of PA-provability.

The supplied NEG18 macro is an instructive negative comparison: it reduces 3,673 source instructions to 3,441, but its best audited physical allocation costs 197 states. It loses useful source sharing and contextual folds. Likewise, a small source DAG does not imply a small blank-start machine when an extra scanner, return control, or initializer must be included.

3 The exact new physical construction

We retain the primary PA194 grammar and all seven folds. Before folding, the new machine has 201 controls: the same 67 reader nodes and 134 fixed controls, with six fixed transitions modified. Native labels are used in this section because they identify inherited control roles directly. Native start and halt are 74 and 254; the exported table is a deterministic breadth-first renumbering with start 0 and halt 194.

Native state	Read	Old action	New action
38	0	$0R28$	$0R11$
38	1	$1R20$	$1R122$
164	1	$1R98$	$1R130$
214	0	$0R188$	$0R312$
214	1	$1R312$	$1R20$
312	0	$0R11$	$0R188$

An action wDq means write w, move in direction D, and enter q. The exact source substitutions are

$$\texttt{node163} = (\texttt{node6}, \texttt{node147}),$$
$$\texttt{node170} = ((\texttt{node36}, (\texttt{node167}, \texttt{node145})), \texttt{node57}).$$

All 190 AX tokens, their order, parameters, and guard remain unchanged. The complete resolved 3,649-site instruction graph equals the supplied 195C graph, including every branch destination. Equality of an expanded word alone would not establish this fact, because structured jumps depend on address geometry.

3.1 The source tail and changed command effects

With zero-based source ordinals, the MP-success tail begins at $E = 3634$, its fetch return is $F = 3637$, and Shift preparation begins at $S = 3638$. The complete 15-instruction tail is supplied in the build receipt. Its MP loop transfers the successful consequent from D to H while clearing D; its Shift loop first discards T, copies H to T and scratch, restores H, and enters the inherited equality-to-one guard. The rest of the source prefix is unchanged except for three retargeted exits at ordinals 48, 3553, and 3633. These are checked by explicit ordinal relocation.

For a command store (t, w, d, h), the new Shift halts if and only if $h = 1$. Original Shift tested $t + h = 1$. For example, $(t, h) = (1, 0)$ causes an original halt but continues under the new command; $(t, h) = (2, 1)$ does the reverse. The command compilers account for this change. Successful MP now returns its consequent in H, clears T and D, and leaves the stack unchanged. Original MP returned the consequent

in T, retained its accumulated D value, and pushed a zero. Thus PA194C changes the actual proof-search controller.

3.2 Compatible history folds

The seven retained native pairs are

$$(2016, 2052),\ (2018, 2027),\ (2022, 2032),\ (2024, 2046),\ (2037, 2015),\ (2055, 2019),\ (2060, 2014),$$

where the first member is redirected to the second. An incoming raw transition to a first member writes 0 and moves right; an incoming transition to its partner writes 1 and moves right. None of these states is initial. Hence the left neighbor identifies the incoming role when a decoder action needs it. The retained state can move left, dispatch through an already available reader, return, and complete the old action.

For each pair and each current read symbol, the local certificate checks both possible predecessor bits. It verifies restoration of every touched cell, the exact final head coordinate, and the projected successor control. All 28 cases are checked in the *final simultaneously folded table*, so a helper affected by another fold cannot silently invalidate a case. The incoming-row census has 23 rows. Each local simulation has positive length, and only the current cell and its immediate left neighbor are inspected. Arbitrary outside-tape extensions therefore preserve the proof. Every other row commutes with the control projection in one step.

Lemma 3.1 (Local quotient transfer)**.** *On every raw run satisfying the incoming-history invariant, one raw step has a finite nonempty simulation in PA194C with exactly the projected control, original head position after that step, and original entire tape. The invariant holds at the initial configuration and at every relevant raw entry. The construction and its correctness are provable in PA.*

Proof. Nonfolded controls use their unchanged action with its target projected. For a folded control, its immediately preceding raw transition supplies the history bit, and one of the finitely many certified two-cell traces applies. Each trace is checked by direct substitution in the literal table. All outgoing raw steps again meet the same incoming-row condition when they enter a folded role. Induction on the raw trace composes these finite positive blocks. All premises are finite table facts, so PA proves them by evaluating their closed arithmetic encodings and then using induction on the trace length. □

The new controller edits do not touch the AX reader states, but this alone would not justify reusing fold certificates: the complete final table is still reconstructed and all local cases are rerun. The resulting count is

$$\boxed{134 + 67 - 7 = 194.}$$

3.3 Universal whole-tape source contracts

A clean source frame contains the current source path, an affine zero gap before its header, nine arbitrary natural-valued unary registers, and an arbitrary finite nonempty spare list. The frame domain has header distance $48 + n$ for arbitrary $n \in \mathbb{N}$; after source path p, the gap is $48 + n - |p|$. The two outcomes of each decrement are treated separately. The contract changes only the prescribed register, source path, and any specified frame displacement. It compares the complete tape and head.

The builder's symbolic register implementation checks all 4,749 branch contracts. A second checker uses literal tape runs and four inductively proved list traversals, rederived from the actual rows of the delivered machine. It does not call an arithmetic-register oracle. Both source graph resolvers independently agree on every instruction destination. Every nonhalt outcome has a finite positive-duration physical simulation; the independent check's duration polynomial has nonnegative coefficients and a positive constant term.

Quantity	Construction check	Independent check
Complete source branch outcomes	4,749	4,749
Symbolic proof transitions	1,507,397	1,096,302
Maximum symbolic transitions per case	506	403
Minimum positive duration constant	3 (independent certificate)	
Generic list lemmas rederived from rows	–	4

These symbolic counts measure finite proof work. They do not bound the literal running time for unbounded unary register values. The induction lemmas and symbolic affine expressions supply the universal quantifiers; a finite sample of small register values would not.

3.4 Exact blank initialization

Starting with the literal blank configuration, 57,922 transitions reach native control 84, head coordinate -2, source site 25, logical register vector $(2, 0, 0, 0, 0, 0, 0, 0, 0, 0)$, and 26 spare blocks of value one. The header distance is 48 (the arbitrary gap offset n is zero). The final tape has 66 one-cells; both checkers compare the complete finite support and assert zero on the entire exterior. The exact one-cell coordinates are in `construction/independent/blank_seed.json`.

This startup is 232 transitions shorter than primary PA194's 58,154-step startup. It is an initialization result only. There is no claim of a uniform speed improvement for arbitrary proofs, source states, or inputs.

4 The calculus, the changed commands, and the PA equivalence

This section gives the ordinary mathematical proof for the literal machine M_{194C}. The numerical calculus and its diagonal calibration come from the supplied PA194 research packet; the changed command protocol comes from the supplied PA195 Combined construction. The new result is their composition with the 194-state table and the fresh source and tape certificates. The detailed inherited arguments are retained in `construction/proof_sources/`; in particular, `original_pa195_calibration.tex`, `diagonal_calculus.tex`, and `combined_commands.tex` identify the proof interfaces being reused. The present argument is a proof of a PA-provability statement. An end-to-end Lean or Rocq theorem for M_{194C} is not claimed.

4.1 A completely specified numerical calculus

Let

$$p(a,b) = \frac{(a+b)(a+b+1)}{2} + b, \qquad \begin{aligned} I(A,B) &= 4p(A,B), & M(a,b) &= 4p(a,b)+1, \\ U(a,B) &= 4p(B,a)+2, & E(a,b) &= 4p(a,b)+3. \end{aligned}$$

Every natural number is a formula code. Code 0 is truth; code $1 = M(0,0)$ is the designated contradiction. The intended readings of I, M, U, E are implication, membership, universal quantification, and equality. Notice the transposition in U. Write

$$N(A) = I(A,1), \qquad A \wedge B = N(I(A, N(B))), \qquad \exists_a B = N(U(a, N(B))).$$

In displays of coded formulas, $A \to B$ abbreviates $I(A,B)$ and $\neg A$ abbreviates $N(A)$. These are operations on natural numbers, not additional primitive syntax. The exceptional identity $I(0,0) = 0$ will be respected throughout.

For arbitrary natural parameters D, A, H, the six head schemas are

$$\begin{aligned}
L_1(D,A,H) &= (\exists_D U(D,H)) \to H,\\
L_2(D,A,H) &= U(D,U(A,H)) \to U(A,U(D,H)),\\
L_3(D,A,H) &= M(H,D) \to (E(H,A) \to M(A,D)),\\
L_4(D,A,H) &= E(H,D) \to (M(A,H) \to M(A,D)),\\
L_5(D,A,H) &= \exists_D E(H,D),\\
L_6(D,A,H) &= U(D,A) \to \big(U(D,A \to H) \to U(D,H)\big).
\end{aligned}$$

There is also the restricted implicational schema

$$\rho(a,b,c,u) = ((c \to u) \to a) \to ((a \to c) \to (b \to c)), \qquad u \le a,$$

and the unconditional diagonal schema

$$\Gamma(B) = B \to U(B,B).$$

Define the pointed-separation expressions

$$\begin{aligned}
V(a,b,c) &= M(a,b) \to U(c,M(a,b)),\\
P(a,b,c) &= c \to M(c,b),\\
Z(a,b,c) &= U\big(b,(M(a,b) \to P(a,b,c)) \wedge (M(c,b) \to c)\big),\\
Q(a,b,c) &= U\big(c,Z(a,b,c) \to \neg P(a,b,c)\big),\\
G(a,b,c) &= \neg\big(V(a,b,c) \to Q(a,b,c)\big).
\end{aligned}$$

The gate schema is $G(a,b,c)$ for $c \neq a, b$. There are exactly two inference rules:

$$\frac{A \qquad I(A,B)}{B} \quad \text{and} \quad \frac{B}{U(a,B)} \quad (a \in \mathbb{N}).$$

The second is generalization of a theorem, not generalization under arbitrary undischarged assumptions. These schemas and rules define $\mathcal{C}_\Gamma$. No equality symmetry, equality substitution, arbitrary formula substitution, or propositional-tautology oracle is primitive. Truth has a finite derivation because $\rho(0,0,0,0) = 0$ satisfies the dominance condition.

The retained PA196 implicational development supplies uniform finite derivations of the classical implicational templates from the restricted ρ schema. Thus the propositional reasoning below abbreviates explicit proof compositions; it does not add a rule to $\mathcal{C}_\Gamma$. The relevant inherited source is `PA196/Logic/ImplicationBasis.lean` in the pinned upstream snapshot, with its arithmetic-certificate counterpart.

4.2 Why this calculus has the consistency strength of PA

We state both syntactic directions, since external truth in a structure would not establish the requested internal equivalence. Fix a standard recursive proof predicate $\mathrm{Prf}_{\mathrm{PA}}$ for first-order PA with full induction, and a recursive predicate $\mathrm{Prf}_\Gamma(d,B)$ for the finite derivation trees just specified.

Lemma 4.1 (Uniform logical calibration)**.** *There are primitive-recursive proof transformations F, J whose correctness is provable in PA and which satisfy*

$$\begin{aligned}
&\mathrm{PA} \vdash \forall p\,\big(\mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner) \to \mathrm{Prf}_\Gamma(F(p),1)\big),\\
&\mathrm{PA} \vdash \forall d\,\big(\mathrm{Prf}_\Gamma(d,1) \to \mathrm{Prf}_{\mathrm{PA}}(J(d), \ulcorner 0 = 1 \urcorner)\big).
\end{aligned}$$

Proof. For the forward direction, the inherited PA196 refutation compiler uses primitive equality through three derived interfaces: two-way membership rewriting, neighborhood equality, and an equality witness. The only uses of its removed equality-vacuity schema in the relevant logical and arithmetic-certificate derivations are the reverse-rewriting and neighborhood modules. Here are the replacements, including the freshness condition that makes them legitimate.

Every variable label occurring in a decoded formula of code B is strictly below B. Indeed, the fields of any nonzero tagged constructor are strictly smaller than its code; induction through its formula children proves the claim, and code zero has no labels. In particular x_B does not occur in its own decoded formula. The gate supplies both membership vacuity $V(a,b,c)$ and $\neg Q(a,b,c)$ by the classical implicational templates.

For reverse rewriting, first let $a \neq b$ and take $\chi(t) = M(d,t)$ or $M(t,d)$ with $d \neq a$, or take $\chi(t) = M(t,t)$. Put

$$C = E(a,b) \to (\chi(b) \to \chi(a)), \qquad \varphi = \neg C, \qquad c = \ulcorner \varphi \urcorner.$$

The code bounds make c distinct from every displayed variable label. Apply the gate at (a,a,c) and abbreviate its Z, P, Q accordingly. Its second separation conjunct and the propositional consequence $\varphi \to \neg\chi(a)$ give

$$Z \to U(a, M(c,a) \to \neg\chi(a)).$$

The equality witness, direct membership rewriting, and membership-only vacuity yield the negative-specialization consequence

$$Z \to (M(c,b) \to \neg\chi(b)).$$

More explicitly, this is the retained specialization using the witness $\exists_a E(b,a)$ and the vacuity of $M(c,b) \to \neg\chi(b)$ at a. Its side conditions are exactly $a \neq b, c$ and, in the nondiagonal cases, $a \neq d$; it needs no equality-vacuity premise. Under φ, P supplies $M(c,a)$ and $E(a,b)$ supplies $M(c,b)$, contradicting the other consequence $\chi(b)$ of φ. Hence $\varphi \to (Z \to \neg P)$. Generalization at c, quantifier distribution, and $\Gamma(\varphi)$ give $\varphi \to Q$. The gate's $\neg Q$ and double-negation elimination prove C. For $a = b$ the result is propositional. If the fixed label $d = a$, direct rewriting first reaches $M(b,b)$, and the already proved diagonal case reaches $M(a,a)$. Thus both membership arguments can be rewritten in either direction, without any remaining label restriction.

For neighborhood equality, let

$$R(a,b,z) = U(z, M(a,z) \leftrightarrow M(b,z)) \wedge U(z, M(z,a) \leftrightarrow M(z,b)),$$

where the biconditionals are the corresponding coded conjunctions. At $z_0 = \ulcorner E(a,b) \urcorner$, the diagonal axiom $E(a,b) \to U(z_0, E(a,b))$ and the rewrites give $E(a,b) \to R(a,b,z_0)$. For $z \neq a, b, z_0$, let $B(t)$ be either membership-only matrix in R. The rewrites, the witness $\exists_{z_0} E(z, z_0)$, and vacuity of $B(z)$ at z_0 give

$$U(z_0, B(z_0)) \to B(z).$$

Its antecedent is vacuous at z: membership vacuity extends through the Boolean connectives and through the outer quantifier by L_2. Generalization and distribution therefore give $U(z_0, B(z_0)) \to U(z, B(z))$. The case $z = z_0$ is already done. This proves $E(a,b) \to R(a,b,z)$ whenever $z \neq a, b$; existential monotonicity and L_5 give the required neighborhood witness. These are precisely the three interfaces of the inherited compiler. Replacing its two affected proof templates gives F. All code choices, label comparisons, and proof concatenations above are primitive recursive; the inherited arithmetic proof construction checks the replacements by the same finite syntactic inductions.

For the reverse direction, work in a fixed conservative definitional extension of PA with 2^z, and set

$$\mathrm{Bit}(s,z) \equiv \exists u, v\, [s = (2u+1)2^z + v \ \wedge \ v < 2^z].$$

Translate $M(a,b)$ to $\mathrm{Bit}(x_a, x_b)$, $E(a,b)$ to $x_a = x_b$, $U(a,B)$ to $\forall x_a\, T(B)$, and 0 to $0 = 0$. For implication, the constructor property is the provable equivalence

$$T(I(A,B)) \leftrightarrow (T(A) \to T(B));$$

when $A = B = 0$, it uses the fixed truth equivalence rather than a false assertion of syntactic constructor disjointness. Bound auxiliary variables in Bit are renamed before substitution. PA proves $\mathrm{Bit}(s,z) \to z < s$, hence $\neg T(1) = \neg\,\mathrm{Bit}(x_0, x_0)$.

The six logical heads have uniform first-order proof templates, ρ is a classical tautology, and the code-freshness argument supplies $T(\Gamma(B))$. The substantive axiom supplier is the gate. For any arithmetic formula $\theta(z, \vec{v})$, PA proves

$$\forall n\, \exists s < 2^n\, \forall z < n\, \big(\mathrm{Bit}(s,z) \leftrightarrow \theta(z,\vec{v})\big). \tag{1}$$

Induct on n: use $s = 0$ at zero, and retain s or replace it by $s + 2^n$ according as $\theta(n, \vec{v})$ fails or holds. This is one fixed proof pattern with a full PA induction instance. Inserting an arbitrary formula code into that pattern and checking the resulting proof are primitive-recursive syntax operations; PA verifies their correctness. This supplies a uniform proof generator, not a uniform truth predicate for arithmetic.

For the gate at $c \neq a, b$, put $\phi = T(c)$. If $a \neq b$, use (1) with bound $x_a + x_b + 1$ and matrix

$$(\mathrm{Bit}(x_a, z) \vee z = x_b) \wedge \phi[z/x_b].$$

Its witness gives $T(Z) \wedge T(P)$, and thus $\neg T(Q)$. If the labels $a = b$ coincide, the first antecedent inside the binder is $\mathrm{Bit}(z,z)$ and is false. The correct matrix is then $(z = x_b) \wedge \phi[z/x_b]$, with bound $x_b + 1$. This separate case prevents treating a bound occurrence of x_a as free. Code freshness ensures that the separator variable x_c does not occur in ϕ. The membership-vacuity clause has its ordinary first-order proof, so $T(G)$ follows.

Recursively translate a numerical derivation: insert these axiom suppliers, concatenate at modus ponens, and append generalization at its corresponding rule. PA verifies the proof-code invariant by induction on the derivation. At conclusion 1, append the fixed proof of $\neg T(1)$ and eliminate the fixed definitional extension. This gives J. The induction is on syntax and proof checking, so its quantified assertion applies to internal, possibly nonstandard proof codes as required. □

4.3 Exact semantics of the Combined commands

Write $[a;w] = p(a,w)$ and $[a_1, \ldots, a_k; w]$ for repeated pairing with a_1 first. The total Cantor projections obey

$$\mathrm{hd}[a;w] = a, \quad \mathrm{tl}[a;w] = w, \quad \mathrm{hd}(0) = \mathrm{tl}(0) = 0.$$

In particular $[0;0] = 0$. All following identities are identities of natural numbers; they do not assume a free list constructor.

A command store is $\langle t, w, d, h\rangle$, with two visible formula codes t, h, a stack w, and an offset d. For $y = d + x$, the new commands are

$$\mathsf{S}\langle t,w,d,h\rangle = \begin{cases} \mathsf{halt}, & h = 1, \\ \langle h, \mathrm{tl}(w), d, \mathrm{hd}(w)\rangle, & h \neq 1, \end{cases}$$

$$\mathsf{MP}_x\langle t,w,d,h\rangle = \begin{cases} \langle 0, w, 0, y\rangle, & I(t,y) = h, \\ \langle 0, w, y, 0\rangle, & I(t,y) \neq h. \end{cases}$$

Only S can halt. Neither branch of MP pushes an entry.

Define $K(a,b,c) = G(a,b,c)$ when $c \neq a, b$ and $K(a,b,c) = 0$ otherwise. For $D = d + x$, $H = h + y$, and $A = c$, AX is exactly

$$\mathsf{AX}_{c,x,y}\langle t, w, d, h\rangle = \langle K(H, D, c), [q_1, \ldots, q_9, t; w], 0, 0\rangle,$$

where, writing $L_i = L_i(D, A, H)$,

$$(q_1, \ldots, q_9) = (\Gamma(A), L_6, L_5, L_4, L_3, L_2, L_1, \rho(D, A, H, d), U(d, h)).$$

The total number of pushed entries is ten, including the saved top. The first eight generated entries and the gate have residue zero modulo four; the last entry has residue two. None is code one. The separately pushed $\Gamma(A)$ is present even when the gate guard fails.

4.4 A proof compiler valid for every incoming stack

Command words execute from left to right. The following exact identities hold for every incoming store, including $t = 1$:

$$\mathsf{AX}_{c,x,y}; \mathsf{S}^9 : \langle t, w, d, h\rangle \longmapsto \langle \rho(d + x, c, h + y, d), [t; w], 0, U(d, h)\rangle,$$
$$\mathsf{AX}_{c,x,y}; \mathsf{S}^{10} : \langle t, w, d, h\rangle \longmapsto \langle U(d, h), w, 0, t\rangle.$$

The first Shift tests the cache cleared by AX; the next nine test the generated entries, never the saved top. The residue calculation proves that all these Shifts continue.

Define

$$\mathsf{Push} = \mathsf{AX}_{0,0,0}; \mathsf{S}^9; \mathsf{MP}_0,$$
$$\mathsf{Pop} = \mathsf{AX}_{0,0,0}; \mathsf{S}^{11},$$
$$\mathsf{Discard} = \mathsf{MP}_1; \mathsf{AX}_{1,0,0}; \mathsf{S}^{10}; \mathsf{MP}_0.$$

Their complete contracts are

$$\mathsf{Push}\langle t, w, d, h\rangle = \langle 0, [t; w], 0, 0\rangle,$$
$$\mathsf{Pop}\langle t, [a; w], d, h\rangle = \begin{cases} \mathsf{halt}, & t = 1, \\ \langle t, w, 0, a\rangle, & t \neq 1, \end{cases}$$
$$\mathsf{Discard}\langle t, w, 0, 0\rangle = \langle 0, w, 0, 0\rangle.$$

Push's final MP fails because it compares residues zero and two. Pop's final Shift tests exactly the saved top. Discard's first MP fails since $I(t, 1) > 0$; the AX identity then leaves $U(1, 0)$ in the top and zero in the cache, so its last MP fails too. These arguments include $t = w = 0$ and the alias $[0; 0] = 0$.

Let Offset_0 be empty and $\mathsf{Offset}_u = \mathsf{MP}_u$ for $u > 0$. On a store with $t = d = h = 0$, it sets d to u and preserves the stack. Then

$$\mathsf{Gen}_i = \mathsf{Push}; \mathsf{Offset}_i; \mathsf{S}; \mathsf{AX}_{i,0,0}; \mathsf{S}^{10}$$

takes $\langle a, w, 0, 0\rangle$ to $\langle U(i, a), w, 0, 0\rangle$, including $a = 1$. The Shift preceding AX tests cache zero; AX clears the retrieved a before its subsequent Shift tests.

For $1 \leq j \leq 7$, the word

$$\mathsf{Discard}; \mathsf{AX}_{A,D,H}; \mathsf{S}^{j+1}; \mathsf{Pop}^{9-j}$$

returns $\langle q_j, w, 0, 0\rangle$ from $\langle t, w, 0, 0\rangle$. After the Shifts, the top is q_j, the cache is q_{j+1}, and the stack is $[q_{j+2}, \ldots, q_9, 0; w]$. There are $9 - j$ displayed entries to consume; each Pop preserves the safe top $q_j \neq 1$, and the last consumed entry is zero. Thus the diagonal selector is explicitly

$$\mathsf{Discard}; \mathsf{AX}_{B,0,0}; \mathsf{S}^2; \mathsf{Pop}^8.$$

It works at $B = 0$ and with a failed gate guard. For $u \le a$, the propositional selector

$$\mathsf{Discard}; \mathsf{Offset}_u; \mathsf{AX}_{b,a-u,c}; \mathsf{S}^9; \mathsf{Pop}$$

returns $\rho(a, b, c, u)$. The gate selector $\mathsf{Discard}; \mathsf{AX}_{c,b,a}; \mathsf{Pop}^{10}$ returns $K(a, b, c)$. All these selectors preserve the arbitrary underlying stack and clear d, h.

For modus ponens the exact suffix is

$$\mathsf{MPSuffix}_b = \mathsf{S}; \mathsf{S}; \mathsf{MP}_b; \mathsf{S}.$$

On $\langle 0, [a, I(a, b), 0; w], 0, 0\rangle$ it halts if $a = 1$ or $b = 1$ and otherwise returns $\langle b, w, 0, 0\rangle$. The first Shift retrieves a; the second tests a and puts $I(a, b)$ in the cache; MP puts b in the cache; the final Shift tests b and consumes the dummy zero. That zero is necessary to preserve w. The same calculation applies when $a = b = w = 0$.

Lemma 4.2 (Uniform compiled refutations)**.** *There is a primitive-recursive compiler $p \mapsto W_p$ such that, for every $\mathcal{C}_\Gamma$ proof p of B and every t, w, W_p from $\langle t, w, 0, 0\rangle$ halts or returns $\langle B, w, 0, 0\rangle$. A proof of 1 yields a command word that halts from every incoming store. PA proves these assertions about the coded words and derivations.*

Proof. Compile axioms by the selectors and generalization by $W_{p_0}; \mathsf{Gen}_i$. For an MP node with minor proof p_0 of a and major proof p_1 of $I(a, b)$, use

$$\mathsf{Discard}; \mathsf{Push}; W_{p_1}; \mathsf{Push}; W_{p_0}; \mathsf{Push}; \mathsf{MPSuffix}_b.$$

The first two words introduce the dummy zero, and the premise computations and Pushes establish the suffix's three-entry precondition. Induction on the proof gives the claimed invariant; an early halt remains a halt after extending the word. If p proves 1, then

$$\mathsf{AX}_{0,0,0}; W_p; \mathsf{Pop}$$

halts from every incoming store. AX first clears d, h; the compiler accepts its arbitrary resulting top and stack; and the final Pop detects top one if execution has not halted earlier. The proof consists of the displayed total projection identities and induction on a finite derivation, so PA verifies the compiler. □

4.5 Canonical stack annotations and the exhaustive search

For $w > 0$ one has $\mathrm{tl}(w) < w$. Thus

$$\delta(w) = \min\{k \le w : \mathrm{tl}^{'k}(w) = 0\}$$

is a total primitive-recursive function, with its defining property provable in PA. Annotate the $\delta(w)$ live heads by proofs, and annotate both t and h. The tail beyond those live heads supplies the fixed truth proof. Pushing a prepends its proof unless $a = w = 0$, in which case the canonical live stack remains empty. Popping zero supplies truth and leaves the empty annotation. This is the exact treatment of the Cantor zero alias.

With either numerical proofs or their PA translations as annotations, AX preserves the invariant: its saved top is known, its heads are axioms, $d \le d + x$ justifies its restricted ρ instance, and $U(d, h)$ follows by generalization. A successful MP uses the annotated t and $I(t, d + x) = h$ to prove $d + x$; a failed MP returns truth. Shift moves annotations or halts with a proof of 1 in the cache. The arbitrary offset d has no theoremhood requirement. Every update is primitive recursive, and PA proves preservation through every coded finite command sequence.

The retained scheduler exhaustively decodes candidate command words. Its inherited coding and padding lemma makes every finite word available at a candidate number above any fixed starting cutoff, including the clean seed's counter 2. Its store is carried between trials. Lemma 4.2's universal incoming-store property is therefore essential: a compiled refutation halts when its trial is reached, unless an earlier trial has already halted. The candidate induction, word decoding, and terminating arithmetic macro routines have primitive-recursive witnesses with PA-verifiable correctness, as in the inherited PA196 source compiler. Conversely, the annotation invariant propagates from the clean store through all prior trials and turns any halting command into a refutation. Together with Lemma 4.1, this proves in PA that the Combined source halts exactly when PA has a refutation.

4.6 From the source to the exact 194-state table

The finite source check establishes equality of the entire 3,649-instruction control-flow map with the established 195C source, under primitive-site ordinal identification. The comparison includes every opcode, register operand, and successor, including both decrement branches. It is stronger than equality of the flattened AX word. The first 3,634 instructions match the original PA194 source except for exit retargetings at ordinals $48, 3553, 3633$. The new MP-success tail begins at 3634, the return to fetch is 3637, and Shift begins at 3638.

The MP-success loop transfers $(D, H) = (y, 0)$ to $(0, y)$ by a decrement, increment, and backward jump, then returns to fetch. The Shift tail clears T, copies H to T and scratch X, and restores H from X. With the inherited $X = 0$ boundary invariant, it reaches $(T, H, X) = (h, h, 0)$ in exactly $2t + 7h + 3$ primitive steps; two more instructions set $X = 1$ and enter the unchanged guard. Induction on the respective unary counters proves these contracts for all natural store values. No AX formula, parameter assignment, or stack order changes.

Fresh physical checks establish all 4,749 source outcomes against the final 194-state table. In particular, the independent literal-tape checker rederives four generic list-sweep lemmas from that table and uses no arithmetic-register oracle. Its frames quantify over arbitrary natural registers, an affine arbitrary gap, and an arbitrary finite nonempty spare list. Each nonhalting source step has a finite physical run to its exact next complete frame; every duration is positive. Increment appends a spare block, while decrement preserves the file length, so the nonempty-spare invariant persists from initialization. The complete blank run reaches the clean site-25 frame after 57,922 transitions, at native state 84, head -2, logical registers $(2, 0, 0, 0, 0, 0, 0, 0, 0)$, and 26 spare registers of value one. The full tape is compared with the specified frame.

These facts admit a specific arithmetic internalization. A literal symbolic transition is a finite table lookup. A compressed unary scan is proved by induction on its run length. Each generic list sweep is proved by induction on its blocks and finite list. The remaining symbolic operations only reexpress the same tape. Their trace constructors, frame encodings, and concatenations are primitive recursive, with correctness proved by these inductions in PA. The machine-specific part is a finite collection of closed table equalities and symbolic traces. The Python checks verify that finite data; they do not themselves constitute a formal PA proof checker.

Theorem 4.3 (PA calibration of the literal PA194C table)**.** *Let H_{194C} be the standard arithmetical sentence asserting a finite halting trace of* `construction/output/pa194C.tm` *from its designated blank start. Then*

$$\boxed{\mathrm{PA} \vdash (\neg H_{194C} \leftrightarrow \mathrm{Con}(\mathrm{PA})).}$$

Proof. A PA refutation gives a numerical refutation by Lemma 4.1, then a uniformly halting command word by Lemma 4.2. The exhaustive scheduler and arithmetic macro contracts give a finite source halt.

Concatenate the exact startup and the physical source-step traces to obtain H_{194C}. All transformations and correctness inductions are available in PA.

For the reverse direction, reason in PA under the temporary assumption Con(PA). The translated annotation invariant precludes a halting command and supplies an annotated successor store at every command boundary. For any T, primitive recursion constructs $T+1$ nonhalting command segments and their positive source and physical simulations. Including startup, this is a physical run longer than T with no halt before a subsequent nonhalt frame. A purported first halting trace of length T contradicts determinism: with partial semantics it cannot be extended beyond the halt, and with absorbing halt semantics it cannot subsequently reach a nonhalt frame. A halt within startup is excluded by its exact finite trace. No inverse decoding of an arbitrary intermediate tape or unbounded search for a return time is needed. Thus PA proves $\mathrm{Con(PA)} \to \neg H_{194C}$. Classical contraposition of the already proved refutation-to-halt implication gives the other direction. □

The full-induction supplier in (1) explains why one cannot replace the searched calculus by the concise iPRA calculus and retain this conclusion without a new calibration. The iPRA project contributes the strategy of replacing expensive primitive schemas by smaller derived interfaces; it does not provide an identification of Con(iPRA) with Con(PA). Likewise, the existing conditional iPRA machine endpoints in the repository do not prove the two machine-specific reductions needed for an unconditional iPRA theorem. The theorem above has the ordinary mathematical PA-provability scope stated at the start.

5 Search below 194 and what the negative results mean

The strict target was a verified machine with at most 193 working controls. The investigated families were chosen from the actual state accounting: reduce the 134-control fixed part, reduce the decoder's raw node count, increase its simultaneously compatible fold count, or change the source and its macro interface while preserving its calibration.

Three independent grammar runs each used a 900-second budget. The following counts are proposals and accepted moves, not distinct grammars or fully certified machines. A score of 60 effective readers gives $134 + 60 = 194$ controls; the sought score was at most 59.

Grammar search	Proposals	Accepted	Best score	Hits < 60
Occurrence rotations	8,830,776	508,531	60	0
Opened copy macros	8,026,437	434,245	60	0
Closed subblocks (corrected)	11,020,325	452,037	60	0

The exact random seeds, elapsed times, inputs, and complete logs are in `search/summary.json` and `search/grammar/`. Proposal counts are machine-speed dependent; the stored seeds and time budgets specify the replay. The three completed runs made 27,877,538 proposals in total.

The local physical experiments used radius-four windows with detours up to 100 transitions, and radius-six windows with detours up to 200. Source-contract contexts supplied 8,338 radius-four windows and 29,347 radius-six windows. The independently derived literal-tape contexts supplied 10,377 radius-four windows across all 4,780 primary-PA194 contracts; eight generic list-sweep controls retained unrestricted contexts. Pair merging, bounded future joins, and incoming-transition elimination returned no accepted candidate. The PA194C register-layout search includes the identity, all 36 transpositions, and seeded random permutations; its last complete progress record is iteration 1,500. Its interrupted tail is not reported as an exact completed count.

A separate SMT experiment attempted three-reader-to-two-reader gadgets. The restricted-target run completed 164 queries in 180.414 seconds; all 164 timed out, with zero satisfiable and zero unsatisfiable

results. The broader-target run logged at least 120 completed queries, all timeouts, before interruption. These solver runs are *inconclusive*, not exclusions. Their sources and logs are retained with the local search receipts under `search/physical/`.

The generic physical searches considered history windows, local detours, common future configurations, and removal of a state by rewriting its incoming transitions. Symbolic contexts cover complete source contracts, including arithmetic-list behavior where available. Register permutations were also screened. None of those bounded families produced an accepted 193-state table. The constraints and explored budgets are evidence about these searches only; they do not prove a state lower bound.

The grammar searches started from both accepted 194-state allocations. They used whole shared-subtree rotations and individual occurrence rotations, scoring raw reader controls minus simultaneously valid folds. A second search opened the copying macros into push and additive-copy subblocks. A third exposed additional closed subblocks and recognized existing fixed reader pairs. The distinction between source tokens and physical labels is essential: two source subblocks can share a physical control only because their surrounding path bits differ. An early experimental expansion incorrectly conflated such source tokens; it was rejected, corrected to preserve token identities, and excluded from every construction claim. Finalists must pass full source-graph, scanner, startup, and universal-runtime checks, not just a numerical search score.

The search failure leaves the 194-state alternative as the completed result. In particular, the manuscript does not use an absent failed counterexample, bounded blank simulation, or an unverified allocator prediction as a proof of correctness.

6 Useful ideas from ZFC and ConciseIPRA

The linked ZFC work contributes methods which can be transplanted only with their invariants. Its conditional-marker constructions replace a control distinction by information in nearby tape cells; an ambiguous incoming role may be harmless if the current symbol excludes the branch where the distinction is needed. Its accepted K215 line jointly designs command-local parameter reading, command order, and selectors. These ideas motivate the local context searches and the explicit command/decoder composition in PA194C [10, 11].

A selector still costs a state unless an existing reader implements it. Consequently an individually valid role merger need not reduce the total. The ZFC residual-parameter construction also illustrates that eliminating an MP parameter can reduce source administration while increasing the physical allocation. The analogous PA route already occurs in the incoming corpus without a sub-196 result. Its state saving cannot be inferred by comparing source expression lengths [12].

ConciseIPRA offers a different conceptual lesson: choose a small set of primitive proof-checker interfaces and derive ordinary logical structure. Its scalar core and clause variants make administrative choices visible in the complete checker. The optimization metric there is Python tokens, AST nodes, and bytes, not complete binary-TM controls [13]. More fundamentally, searching iPRA proofs does not by itself implement PA-refutation search. The present soundness calibration uses finite binary comprehension for an arbitrary formula, supplied by full PA induction. A replacement theory would require explicit proof translations in both directions and a complete compiler, including proof representation and blank initialization. No claim that Con(iPRA) may simply replace Con(PA) is used here.

7 Novelty check and reproducibility

The new dense graph is compared against all four prior 194-state graph classes, including the A/D variant and the rejected scanner experiment. The comparison allows every initial root reaching the

full graph, arbitrary control renaming, global left/right reflection, and simultaneous read/write bit complementation. All four orientations of PA194C have canonical hashes absent from the old corpus. Symbol complementation is included only to exclude cosmetic novelty; by itself it changes the blank symbol and need not preserve blank-zero semantics.

The unreflected, uncomplemented root-free graph hash is

```
b79f4d344740679b87dace02fe3f211565c2cd4513f346812570d90caa3d9a3f
```

The receipt records all four comparisons. This nonisomorphism is additional evidence for a different machine, while the changed command semantics establish the substantive algorithmic difference. Neither graph novelty nor a smaller source alone establishes the consistency theorem.

From the package root, Python 3.10 or later with its standard library suffices for the primary checks:

```
python3 -B construction/build194c.py
python3 -B construction/verify194c_independent.py
python3 -B construction/command_audit/check_composite_commands.py
python3 -B comparison/compare_candidate.py \
  --candidate construction/output/pa194C.tm
```

Assertions must be enabled; do not use `-O` or set `PYTHONOPTIMIZE`. The complete universal checks are the defaults. The builder's optional skip flag is for diagnostics and is not a substitute for verification. Neither the optional searches nor a proof-assistant installation is needed to reconstruct the machine and replay its finite and symbolic certificates.

The recursive corpus census can be rerun with

```
python3 -B comparison/census_current.py \
  --repo /path/to/ProveIt --output /tmp/pa-census
```

using the pinned repository. Its complete older input archives are not repeated in this ZIP; their exact hashes and member paths are retained. All seven existing 194-state byte forms are included under `comparison/reference194/` for a smaller direct comparison.

To rebuild this report, run `pdflatex` twice on `report/pa194C_conpa.tex` from the report directory. The file `SHA256SUMS` binds all release artifacts except itself. Replaying checks may rewrite receipts, so verify the original manifest first and then regard regenerated receipts as a new local run.

7.1 Proof and trust boundary

There are three distinct kinds of evidence. Literal table census, conjugacy, complete startup, source graph equality, and finite fold traces are finite facts checked by executable programs. Universal unary and list contracts use explicit symbolic identities and inductive loop arguments. The PA consistency equivalence additionally uses the mathematical proof-producing translations and command compiler of Section 4. Python execution is not presented as a proof kernel certifying that entire PA theorem.

The inherited Lean files are provenance and further-formalization references. This release does not claim an end-to-end Lean or Rocq elaboration, and does not promote the repository's conditional iPRA endpoint to a completed theorem. No global minimum state count is proved.

A Complete transition table

Each entry wDq specifies the written symbol, direction, and next state. The four blocks are read independently. Start is 0 and halt is 194; there are no outgoing rows for halt. All unlisted tape cells initially contain zero.

q	0	1	q	0	1	q	0	1	q	0	1
0	1R1	1R2	49	0R68	0R69	98	0R111	1R75	147	0R149	1R158
1	1R3	1R1	50	0R70	0L48	99	0L63	0R109	148	0R159	1R88
2	0R0	1R2	51	0R64	1R53	100	0L62	1R113	149	0R97	1R160
3	0R4	0R1	52	0L71	0L15	101	0L90	1R114	150	0R161	1R162
4	1R5	1L6	53	0R72	0L10	102	0R115	0L14	151	0R163	0L10
5	1L7	1L8	54	0R73	1R74	103	0R69	1R103	152	0R164	1R165
6	0L7	1L6	55	0R75	1R76	104	0R116	1R95	153	0R166	1R157
7	0L9	1L6	56	0R77	1R78	105	0R116	1R67	154	0R167	1R156
8	0L10	0L11	57	0R79	1R80	106	0R117	0L14	155	0R135	1R130
9	1R12	0L9	58	0L81	0R1	107	0R118	1L71	156	0R147	1R41
10	0L9	0L10	59	0R82	1R59	108	0R119	1R120	157	0R134	1R168
11	1L8	1L11	60	0L73	1L46	109	0R14	1R109	158	0R41	1L169
12	0R12	0R13	61	0R83	1R84	110	0R66	1R110	159	0L170	1R171
13	0L14	0R13	62	0R54	1R85	111	0R121	1L35	160	0R107	1R85
14	1L15	0L15	63	0L86	1R87	112	0R122	1R123	161	0R172	1R170
15	0L15	1L16	64	0R58	0R82	113	0R88	0L14	162	0R173	1R174
16	0L17	1L16	65	0R88	1R30	114	0R90	1L95	163	0R175	1R176
17	0L18	1L16	66	0R89	1R66	115	0R124	1R124	164	0R177	1R178
18	0L19	1L16	67	0L90	1L33	116	0L63	0R103	165	0L179	1L180
19	0L20	1L16	68	0L81	0R2	117	0R68	0R91	166	0R149	1R140
20	0L21	1L16	69	0R91	1R69	118	0R58	0R91	167	0R165	1R181
21	0L22	1L16	70	0R72	1R92	119	0L86	1R125	168	0L130	1R88
22	0L23	1L16	71	0R46	1R93	120	0R126	1R127	169	0R182	1R135
23	0R23	1R24	72	0R42	0L13	121	0R58	0R110	170	0R183	1R145
24	0R25	1R26	73	0R94	1R37	122	0R99	1R113	171	0R158	1R130
25	0R27	1R28	74	0R30	1R95	123	0R79	1R78	172	0R139	1R165
26	0R29	1R30	75	0R42	1R96	124	0R88	1R88	173	0R158	1R184
27	0R31	1R194	76	0R97	1R98	125	0R128	1R129	174	0R185	1R107
28	0L32	1R33	77	0R99	1L100	126	0R32	1R36	175	0R31	1R52
29	0R34	1R35	78	0R30	1R77	127	0R31	1R111	176	0R186	1R175
30	0L36	1L37	79	0R93	1R101	128	0R130	1R131	177	0R134	1R142
31	0R38	1R39	80	0R30	1R102	129	0R132	1R133	178	0R32	1R187
32	0R40	1R41	81	1R81	1L15	130	0R134	1R135	179	0R144	1R188
33	0R42	1R43	82	0R103	1R82	131	0R136	1R137	180	0R189	1R190
34	0R44	1R45	83	0R94	0L7	132	0R138	1R139	181	0R41	1R191
35	0R46	1R33	84	0R104	1R105	133	0R140	1R141	182	0R168	1R41
36	0R47	1R48	85	0R42	1R106	134	0R97	1R35	183	0R192	1R88
37	0R49	1R50	86	0R107	1R48	135	0R142	1R124	184	0R153	1R146
38	0R51	1R42	87	0R108	0L72	136	0R143	1R144	185	0R165	1R180
39	0R52	1R53	88	0R68	0R66	137	0R145	1R146	186	0R92	1R111
40	0R54	1R55	89	0R109	1R89	138	0R134	1R147	187	0R193	1R159
41	0R56	1R57	90	0R49	1R100	139	0R88	1R148	188	0R169	1R154
42	0R58	0R59	91	0R110	1R91	140	0R149	1L131	189	0R158	1R192
43	0R60	0L14	92	0R30	1R48	141	0R150	1R151	190	0R179	1R182
44	0R61	1R62	93	0R47	1R60	142	0R76	1R41	191	0R182	1R41
45	0R63	1R52	94	0L63	0R89	143	0R28	1R152	192	0R149	1R168
46	0R64	1R65	95	0R94	1L111	144	0L153	1L154	193	0R76	1R144
47	0R58	0R66	96	0R101	0L14	145	0L129	1R155			
48	0R67	0L14	97	0R36	1R112	146	0R156	1R157			

B All 36 incoming archives

The catalogue records each top-level archive at the pinned revision. Reported "results" below refer to the supplied evidence status, not to independent proof-assistant verification of every historical table. Nested payloads and every recognized machine occurrence are recorded in the machine-readable census.

Archive	Principal result	Contribution and status
Below_U208_Research_2026-10-29.zip	208 baseline	Seven-cell context windows and guarded source pruning; no accepted smaller table.
ConPA_below208_research_2026-10-29.zip	208 baseline	Logical-role obstruction and 279,936 restricted parameter cases; no accepted smaller table.
PA195_ConPA_research_2026-11-14.zip	195	Separate190: 195S has original commands, 195C has Combined commands; 67 readers and six folds.
PA_consistency_TM_research_2026-11-13.zip	196 baseline	Five-zero source geometry, a 157-node compressed source, and parameter-free MP compiler; no physical saving.
U208_ConPA_2026-10-29.zip	208	Original 172/211 history fold, with two rooted variants; source and blank initialization retained.
U208_ConPA_Conditional_Context_2026-10-29.zip	208	Same two U208 graphs and state saving, expressed with conditional entry contexts.
U208_ConPA_Entry_Context_2026-10-29.zip	208	Same U208 fold with complete entry-context and positive-block proof; not an additional saving.
U208_ConPA_History_2026-10-29.zip	208	Same two U208 classes through finite histories; failed further reduction retained.
conpa_tm_comparison_and_reduction_search.zip	196 experiments	Two MP-command variants; two rejected Shift tables actually halt after 43,563 blank steps.
conpa_tm_comparison_and_search.zip	196	Physical A/D register swap; a proposed two-rotation 195 fails 93 primitive contracts.
conpa_tm_study_2026-11-13.zip	196	Combined controller: cached-head Shift and MP result in H; source has 3,788 sites.
pa194_conpa_2026-11-13(2).zip	194	Separate190 with 134+68-8 controls; eight folds, 32 local cases; also an intermediate separate195.
pa194_conpa_2026-11-13.zip	194	Separate190 with 134+67-7 controls; seven folds, 28 local cases; additional A/D layout.
pa195_conpa_2026-11-13.zip	195	Original diagonal191 construction: 134+68-7; 3,700 source sites and written PA calibration.
pa195_diagonal_combined_2026-11-13.zip	195	PA195DC combines diagonal191 with Combined commands; optional rho-permutation and negative-vacuity variants.
pa195_followup_research.zip	195	M195S8 repairs a rejected 194 by adding a scanner control: 135+68-8; exact startup 59,164.
pa196_research_2026-11-13.zip	196 baseline	A 157-node source representation; lost contextual helper prevents the advertised prospective 195 allocation.
pa208_research_2026-10-29.zip	208	Eighteen further helper graphs; 27 designated presentations in 20 U208 rooted classes.
pa_below196_research.zip	196 baseline	Eight equivalent gate forms, orientation changes, and freshness obstruction; no smaller validated allocation.
pa_consistency_comparison_2026_11_13.zip	196	Modal-head/explosion split and twelve semantic compositions; countermodels rule out naive head deletions.
pa_consistency_comparison_and_new196_2026-11-13.zip	196	Diagonal191, separate190, and scanner-best implementations; logical and source ancestors of the smaller tables.
pa_consistency_cross_archive.zip	196	Four diagonal/separate emitter plus Combined-controller compositions, optionally with zerojump.
pa_consistency_reduction.zip	196	Rebracketing, H-only Shift, and larger integrated gates; two proposed 195 allocations fail runtime contracts.

Archive	Principal result	Contribution and status
pa_consistency_reduction_audit_2026-11-14.zip	196 experiments	Further MP-in-H combinations and parameter/helper searches; no supported smaller result.
pa_consistency_research.zip	196	A 158-node source needs an extra scanner; nominal 195 misroots a legal frame.
pa_consistency_research_2026-10-29.zip	208 baseline	Proved source specialization from 3,909 to 3,904 instructions; no physical state saving.
pa_consistency_research_2026-11-13(2).zip	196	Twelve-rotation 196 and diagonal198; last-successful-MP halting is a source-only experiment.
pa_consistency_research_2026-11-13.zip	199 new	Diagonal vacuity reduces AX 201 to 191 items and source 3,812 to 3,700; first implementation 199.
pa_consistency_research_2026-11-14.zip	197 new	NEG18 reduces separate source to 3,441 sites; extra physical cost prevents a state improvement.
pa_tm_comparison_and_reductions.zip	196	Two rotated layouts; equivalent distributed gate shortens the source but compiles to 198.
pa_tm_extended_research_2026-11-13.zip	196	Four closing-decrement choices and stronger invariants; all alternatives touch the same row.
pa_tm_research_2026-11-13(2).zip	196	Invariant-based zerojump; proposed 195 helper fold has wrong successor and register update.
pa_tm_research_2026-11-13.zip	196	Alternative axiom order and shorter source paths; unchanged 3,812-site source CFG.
pa_unary_power_cover.zip	no table	Unary covering theory and uniform finite-proof translations to and from PA.
u208_comparison_research_2026-10-29.zip	208 baseline	Original two-class audit; rejected 207 has a concrete wrong-frame witness.
u208_research_2026-10-29.zip	208	Compatibility-family history fold; same original U208 improvement with complete symbolic contracts.

References

[1] V. Reshetnikov, *ProveIt*, pinned revision `768c2be1a8e4d2fcf8aa3443c7b0f084d145734c`. Repository snapshot.

[2] *PA consistency machines*, repository README and formal-proof index. Pinned PA research index.

[3] *Incoming research packets*, all 36 archives at the same revision. Pinned archive directory.

[4] Supplied *A 194-state Turing machine calibrated to Con(PA)*, `pa194_conpa_2026-11-13.zip`; primary table, grammar, seven-fold quotient, PA calibration, and independent literal-tape checker. See the exact archive SHA-256 in `construction/PROVENANCE.json`.

[5] Supplied *PA194: a 194-state binary Turing machine for Con(PA)*, `pa194_conpa_2026-11-13 (2).zip`; 68-reader, eight-fold construction and ten-entry command audit.

[6] Supplied `PA195_ConPA_research_2026-11-14.zip`, 195S and 195C constructions; `construction195`, `physical195_audit`, and `independent_verification`. The date in the filename is retained from the supplied UTC-named archive. The exact component manifest is in `construction/PROVENANCE.json`.

[7] Supplied original PA195 and diagonal-calculus calibration, preserved under `construction/proof_sources/` with its upstream references and license notices.

[8] James Chang, PA196 and PA209 developments, imported snapshots under `Computability/BusyBeaver/Research/PA/jamesyc-upstream`. Pinned imported sources.

[9] *ZFC consistency research*, pinned subtree.

[10] ZFC conditional-marker alternative, candidate-k215/alternatives/conditional-marker/README.md, within the pinned ZFC subtree.

[11] ZFC `candidate-k215/README.md`, command-local field reading, command reordering, and partial selector construction.

[12] ZFC `k215-residual-parameter/README.md` and `proof.md`, residual-parameter command compiler and physical costs.

[13] *ConciseIPRA*, scalar core and clause presentations, pinned source and reports.